

of the j -th scoring question is r'_{ij} (the same definition as Section 4.2). The results of *Original Test* will be calculated as $\frac{1}{N} \sum_{i=1}^N (\bigwedge_{j=1}^{m_i} r'_{ij})$, and the results of *Coherent Test* will be calculated as $\frac{1}{K} \sum_{k=1}^K \bigwedge_{i \in \text{Group}(k)} (\bigwedge_{j=1}^{m_i} r'_{ij})$. Instructions containing Selection are categorized as either single-layer or multi-layer nested, respectively. As shown in Figure 7, for single-layer Selection instructions, LLMs with stronger instruction-following abilities show a smaller performance drop in the coherent test, which better understands the selection structure. For more complex multi-layer nested Selection instructions, even the state-of-the-art LLM, GPT-4, achieves only 14.9% accuracy in the coherent test. At the same time, smaller-scale LLMs can't perfectly follow any group of instructions. The results highlight current LLMs' weaknesses in following multi-layer tree-structured instructions.

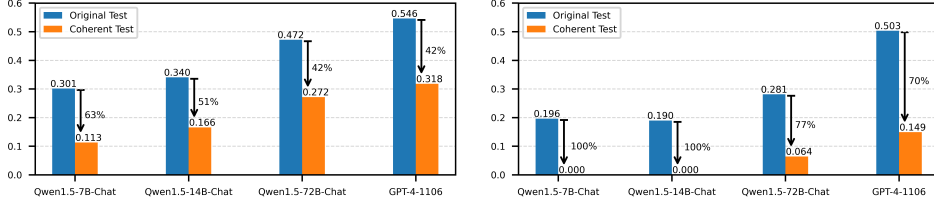


Figure 7: The performance variance under the coherent test for *Selection*. The left side represents single-layer *Selection* instructions, and the right side corresponds to multi-layer *Selection* instructions.

Comparisons between Other Capabilities. We compare the performance of representative LLMs across 3 prominent LLM evaluation benchmarks in addition to COMPLEXBENCH: IFEval[19], HumanEval[48], and MATH[49], focusing on instruction-following, coding, and mathematical ability, respectively. As shown in Table 7, although the performance of various LLMs on COMPLEXBENCH is well correlated with their performance on other benchmarks, the rankings of LLMs on COMPLEXBENCH do not entirely correspond with those on the other three benchmarks. For instance, ChatGLM3-6B-Chat demonstrates outstanding coding and mathematical abilities among LLMs of similar scale, but it notably struggles with complex instruction-following. On the other hand, while Llama-3-70B-Instruct surpasses GPT-4-1106 on IFEval and ranks first, it still shows a performance gap with GPT-4-1106 on COMPLEXBENCH. This discrepancy is primarily in the areas of instructions with complex constraints composition, which are not covered by IFEval, indicating that COMPLEXBENCH can provide a complementary perspective for LLM evaluation.

Model	COMPLEXBENCH	IFEval	HumanEval	MATH
GPT-4-1106	0.800	75.4	84.6	64.3
GLM-4	0.779	66.7	72.0	47.9
Qwen1.5-72B-Chat	0.752	55.8	71.3	42.5
Llama-3-70B-Instruct	0.757	78.9	81.7	50.4
Llama-3-8B-Instruct	0.638	68.6	62.2	30.0
Mistral-7B-Instruct	0.592	40.5	30.5	13.1
Qwen1.5-7B-Chat	0.658	38.8	46.3	23.2
InternLM2-7B-Chat	0.634	46.5	59.8	23.0
ChatGLM3-6B-Chat	0.546	28.1	64.0	25.7
Correlation with COMPLEXBENCH	-	0.814	0.715	0.895

Table 7: Model comparison on different abilities. The last row shows the Pearson correlation between the performance of LLMs in COMPLEXBENCH and other benchmarks.

6 Conclusion

In this work, we propose COMPLEXBENCH, a systematical benchmark for complex instruction-following. We first propose a hierarchical taxonomy for complex instructions, including 4 constraint types, 19 constraint dimensions, and 4 composition types. Furthermore, we manually collect a high-quality dataset accordingly. Along with the dataset, we propose a structure-aware automatic evaluation method for complex instruction-following with constraints composition and further enhance the evaluation accuracy by equipping LLM-based evaluators with rules. Finally, we conduct extensive experiments to evaluate the performance of current representative LLMs on complex instruction-following and uncover their significant deficiencies in dealing with complex composition types. In summary, we posit that COMPLEXBENCH can serve as a valuable tool for benchmarking the complex instruction-following ability of LLMs, providing a complementary perspective for LLM evaluation and useful insights for further work to improve this ability of LLMs.

Acknowledgements

This work was supported by the National Science Foundation for Distinguished Young Scholars (No. 62125604), the NSFC projects (No. 62306160), and the Tsinghua University Initiative Scientific Research Program. We would also like to thank Zhipu AI for sponsoring the computation resources and annotation costs used in this work.

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